

Response to Reviewers

Paper 1217, Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

We thank the reviewers and the SPC for a careful and specific set of comments. Every point is answered below, with what we did and where to find it. Additions are highlighted in blue in the revised manuscript.

Four changes are substantive rather than editorial: the repost reconstruction is validated against an envelope of alternative linking rules (C1, C2); the ideology labels carry class-specific and platform-specific validation plus a noise-propagation analysis (C4, C6); cascades are reported under both a post-level and a user-collapsed definition (C3); and a degenerate scale estimate in the regression models is corrected, with every specification refitted (C12).

The central finding survived every check. Ideological composition and alignment absorb 81 to 94 percent of the baseline platform interaction across two samples and two scale estimators, never below 68 percent under any spline placement, and never below 70 percent in the worst of four label-noise processes. It also holds across ideology thresholds from 0.5 to 0.7, under both composition encodings, and under every repost reconstruction rule.

Two checks qualified a magnitude rather than the conclusion, and we report both rather than only the stronger specification. The user-collapsed representation shows that part of the raw breadth gap comes from repeat posting rather than from more distinct participants, and the center-category check shows that some of the statistical accounting runs through partisan versus non-partisan participation. We have narrowed the claims accordingly.

#	Raised by	Reviewer point	What we did	Where
C1	SPC-1, R1-R3	Repost cascades are reconstructed, not observed, so the cross-platform similarity may be an artifact of the method.	Rebuilt all 244,129 repost cascades under five alternative linking rules plus a 40-draw random-rule ensemble. The platform interaction stays within [-0.012, +0.031] for breadth and [-0.083, +0.028] for depth under every rule, against +0.170 and -0.182 for replies. These cascades are called reconstructed throughout.	App. D, Fig. 9
C2	R3, SPC-1	A user may have followed someone only after reposting, and that error could differ by platform.	Both follow networks were collected at the same time, so any timing gap applies equally to the two platforms. Deleting 5 to 30 percent of follow edges at random moves the breadth interaction from 0.031 to 0.005 and depth from 0.004 to 0.013, an order of magnitude below the reply values.	App. D
C3	SPC-2, R2, R3	Unclear whether cascade nodes are users or posts, and how repeated users in one thread are handled.	Reply nodes are posts; ideology is aggregated per user, so a user who replies twice appears twice carrying the same label. We rebuilt every cascade under a user-collapsed alternative. Depth holds; breadth shrinks, so part of Truth Social’s extra width is repeat posting rather than more distinct participants. Both are reported.	Methods; App. E

#	Raised by	Reviewer point	What we did	Where
C4	SPC-3, R1, R2	Ideology validation is thin: 200 items, no class-specific or platform-specific performance.	Added per-class and per-platform precision and recall with bootstrap intervals against the 171 consensus items. Accuracy 0.86; kappa 0.77 between annotators, 0.61 and 0.74 model to human. Minority-class precision is 0.64 on both platforms against 0.98 and 0.93 for each platform’s largest class.	App. B, Table 3
C5	SPC-3, R2	The center category may mix genuinely centrist, mixed, and uncertain users.	Center is now defined explicitly as a residual category, and we report the models recomputed over partisan users only.	Methods; App. B
C6	SPC-3, R1	Show the findings survive plausible ideology label noise.	Propagated the measured confusion matrix through the full pipeline, 100 draws per scenario across four error processes including a worst case built on the stricter annotator. The median run retains 86 to 96 percent of the attenuation, the worst case 70 percent.	App. B
C7	R1	The left share on Truth Social and right share on Bluesky look too high against prior work.	Our frame is Biden and Trump conversation during the campaign, not the platform population. A prevalence correction lowers the Bluesky right share from 24.2 to 17.9 percent and the Truth Social left share from 11.5 to 1.9 percent. A gap remains for Bluesky, which we attribute to the sampling frame rather than to label error.	App. B
C8	SPC-4, R2, R3	Alignment is computed from the same reply edges it is meant to explain, so causal language is too strong.	Causal language is replaced by statistical accounting throughout, and the simultaneity is stated directly in the Limitations subsection.	Results; Limitations
C9	SPC-5, R1	Readable descriptive statistics are missing.	Added cascade counts, post counts, and the cascade size distribution by platform and cascade type. Compiling it surfaced something we had not reported: most cascades are a single post, and the share differs sharply by platform, 71.3 percent of Bluesky reply cascades against 43.3 percent on Truth Social. Those cascades sit at the origin of the scaling plot, so we refitted every model on the 47,670 cascades with more than one post. The baseline interaction is smaller there, as expected once those points are removed, and Model 3c still absorbs 81 to 92 percent of it.	App. C, Table 4; App. I, Table 6
C10	R1	Report raw motif counts, not only z-scores.	Added observed counts for all 54 ideology-labeled three-node motifs on both platforms.	App. G, Table 5
C11	R1	Motifs overlap; does the null model account for that?	It does: randomized graphs are counted with the identical enumeration, so overlap inflates observed and null counts alike. We now state the procedure precisely, 100 randomizations by within-cascade double-edge swaps with 5m attempts, and note the consequence that neighboring z-scores are correlated and should be read as a pattern across families.	Methods; App. G

#	Raised by	Reviewer point	What we did	Where
C12	SPC-6, R2	Main text says Huber loss, appendix says OLS with HC3. Which is it?	Huber. The stale appendix paragraph is removed. Checking this surfaced a further problem: the default scale estimate degenerates here because 61 percent of cascades sit at the origin, so every model is refitted with Huber’s proposal 2 scale and Table 1 now carries interpretable standard errors. A sensitivity table reports both estimators.	Methods; App. I, Table 6
C13	SPC-7, R1-R3	One month of Biden and Trump content in an unusual period: what generalizes, and what do the two shapes imply?	The Discussion is rewritten so implications come first, followed by a Limitations subsection stating what a candidate-centered, one-month frame does and does not support.	Discussion
C14	R1	Excluding influencers only on Truth Social is poorly justified; why not the top n percent on both?	The follower distributions are not comparable at the top. The largest Truth Social account has 56 times the followers of that platform’s own 99.9th percentile account; on Bluesky the ratio is 5. The top eight Truth Social accounts hold 16 percent of all followers against 6 percent on Bluesky. A symmetric top-n rule would delete ordinary Bluesky accounts to mirror a structure only Truth Social has.	App. A
C15	R1	How many reply chains are two users going back and forth?	Among cascades of depth two or more, 68.8 percent on Bluesky and 70.0 percent on Truth Social contain an A to B to A exchange, carrying 20.0 and 15.6 percent of reply edges. The rate is similar on both platforms, so the depth difference is not explained by dyadic exchange.	App. F
C16	R1	More topic model detail; BERTopic often leaves half of short posts as outliers.	Added the corpus size (125,623 root posts), the outlier reduction step and its settings, the resulting outlier share, and the eleven retained topics.	App. H
C17	R1	The alignment results are hard to follow.	The subsection is rewritten with a plain definition and a worked example, and notation is defined at first use.	Methods
C18	SPC-5, R1	Figures are messy: overlapping labels, unclear quantities.	Rebuilt the motif figures with family labels moved out of the data area, and the topic, partisanship and partial-residual figures, where value labels had been falling outside the axes. All are 300 dpi with no overlapping elements.	Figs. 1, 4, 5, 7, 11
C19	R1	Citations are dated; add recent information-spreading work.	Added recent work on cascades and newer platforms, including the Bluesky diffusion literature.	Related Work
C20	R1	Unexplained notation and stray paragraph indents.	Notation and formatting pass.	Throughout
C21	R1	Another US-centric paper.	Acknowledged in the Limitations subsection, with non-US comparison named as future work.	Limitations